

SUMIT KUMAR

GenAI & Multi-Agent Engineer | Full Stack Engineer

Noida, India | 7011676185 | sumitsumitsumit163@gmail.com | LinkedIn | GitHub | Portfolio

PROFILE SUMMARY

Forward-thinking **Generative AI & Full Stack Engineer** with strong expertise in building scalable Multi-Agent Systems and RAG architectures. Currently specializing in the **IIT Mandi GenAI curriculum**, focusing on LLM orchestration, fine-tuning and agentic workflows using **LangChain and CrewAI**. Proven track record of integrating AI-driven automation into enterprise web applications (MERN/Next.js) to optimize performance and decision-making.

TECHNICAL SKILLS

- **Generative AI & LLMs:** RAG Pipelines, Prompt Engineering (Zero-shot/Few-shot/CoT), LLM Fine-Tuning, Transformers.
- **Agentic Frameworks:** Multi-Agent Systems, LangChain, LlamaIndex, LangGraph, CrewAI, AutoGen, ReAct, MCP.
- **AI Development:** Gemini, OpenAI, Claude, Hugging Face, FastAPI, RESTful APIs, JWT Authentication.
- **Database & Reporting:** Vector DB (Pinecone, ChromaDB, FAISS), MySQL, PostgreSQL, MongoDB, Metabase, SSRS.
- **Full Stack Core:** React.js, Next.js, Python, Pandas, TypeScript, JavaScript, Node.js, Express.js, Tailwind CSS, MUI.
- **Deployment & ML Ops:** Docker, AWS, Jenkins, CI/CD, LangSmith/LangFuse.
- **Tools & Utilities:** Git/GitHub/GitLab, Postman, Jupyter, Vercel.

PROFESSIONAL EXPERIENCE

Software Engineer, Kent RO Systems Pvt Ltd

Apr 2024 — Present

- Engineered an OCR-based **multilingual** invoice parsing pipeline using **AWS Textract** and **Mistral LLM**, extracting unstructured data and cutting manual processing time by **60%**.
- Optimized backend API performance and implemented NGINX load balancing, decreasing system latency by **15%** and ensuring high availability.
- **Integrated** dynamic **Zoho Analytics** dashboards with role-based and criteria-based filters, enhancing decision-making and data accessibility across departments.
- **Developed** secure authentication and **role-based access control** using JWT and OAuth2 for internal and external stakeholders.
- Developed and automated complex enterprise reports using **SSRS and Metabase**, providing real-time data visibility and accelerating cross-departmental decision-making.

Associate Software Developer, Technoidentity

Aug 2023 — Mar 2024

- **Revamped** the HRMS platform using **Next.js and TypeScript**, improving UI performance and overall user experience.
- **Developed** scalable and reusable UI components for the **Hectare (US Toll Authority)** platform, enhancing system reliability and maintainability.
- **Automated** end-to-end test cases using **Cypress and Playwright**, reducing manual QA effort and improving testing efficiency.

PROJECTS

Kent Risengine, Internal

- **Engineered** a multilingual AI invoice parser using **AWS Textract** and **Mistral LLM**, boosting data processing by **40%**.
- **Architected** high-performance components and validation schemas using **React (SSR)** and **Node.js** to ingest data.
- **Integrated** dynamic **Zoho Analytics** and **SSRS** dashboards with role-based filters to expedite data-driven decisions.

Cam Attendance, Internal/Client

- Maintained and scaled a full-stack attendance platform utilized by thousands of concurrent employees.
- Managed REST APIs (Spring Boot) and frontend modules (Angular) to orchestrate real-time HR data tracking and synchronization.
- Integrated automated data pipelines to sync and validate check-ins, leave logs, and key HR metrics.

TIOS Revamp, Internal

- Migrated a legacy HRMS to a modern modular architecture using Next.js.
- Improved performance, maintainability, and scalability by redesigning the UI/UX and restructuring the entire codebase.

KEY ACHIEVEMENTS

- Engineered an AI-driven, multilingual invoice automation pipeline, successfully cutting manual data entry operations by **60%**.
- Boosted system performance and API response times by **15%** through optimized server load handling and refined backend logic.
- Built secure Node.js authentication protocols featuring granular role-based access and integrated AWS certificate handling.
- Strengthened overall platform architecture by establishing modular codebases, improving cross-service data flow and scalability.

EDUCATION

MTech – Computer Science2022 — 2024

Monad University

B.E – Mechanical Engineering2012 — 2016

RGPV

CERTIFICATIONS & TRAINING

PG Certification in Gen AI and Multi-Agent Systems, IIT Mandi - TIHDec 2025 — Present

- **Core Curriculum:** Mastered advanced Prompt Engineering, NLP fundamentals, and Retrieval-Augmented Generation (RAG) pipelines using Vector Databases.
- **Agentic AI & Fine-Tuning:** Architected autonomous Multi-Agent Systems (MAS) using LangChain, CrewAI, AutoGen, LlamaIndex and fine-tuned Large Language Models.
- **Key Learnings & Deployment:** Applied Deep Learning foundations and Transformer architectures to build and deploy scalable GenAI models via FastAPI.
- Deployed **FastMCP** servers to enable secure, standardized tool-calling for autonomous AI agents.
- Utilized **Pandas** for complex data preprocessing within Agentic RAG workflows.

Full Stack Web Developer, Newton SchoolMay 2022 — Mar 2023

- Trained in modern full-stack web technologies with a strong focus on building AI-powered web applications.
- Gained hands-on experience with HTML, CSS, JavaScript, React, Next.js, Node.js, Express.js, MongoDB, and Git.
- Actively participated in DSA coding contests, improving problem-solving and algorithmic thinking.
- Collaborated on group projects and real-world development challenges, strengthening teamwork and practical development skills.